Homework #1 of Introduction to VLSI Design

- 作業繳交截止日期: March 12, 2025, 23:59 (2 week)
  (上傳E3數位平台繳交)

  24 hours after the deadline: 20% penalty
  48 hours after the deadline: 50% penalty
  No submissions accepted after 48 hours

- 檔名 : VLSI_HW1_學號.pdf (VLSI_HW1_112345678)
  請將檔案轉成PDF格式 (命名錯誤扣5分)

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Homework #1

- Use Stixu Stick Diagrammer to draw stick diagram for the following cells
  - (20%) NOR2: not(A+B)
  - (20%) NAND3: not(ABC)
  - (20%) AOI21: not(AB+C)
  - (20%) OAI22: not((A+B)(C+D))
  - (20%) AOI32: not(ABC+DE)

- Requirements
  - Logic correctness
  - Submission
  - Screen dump with signature

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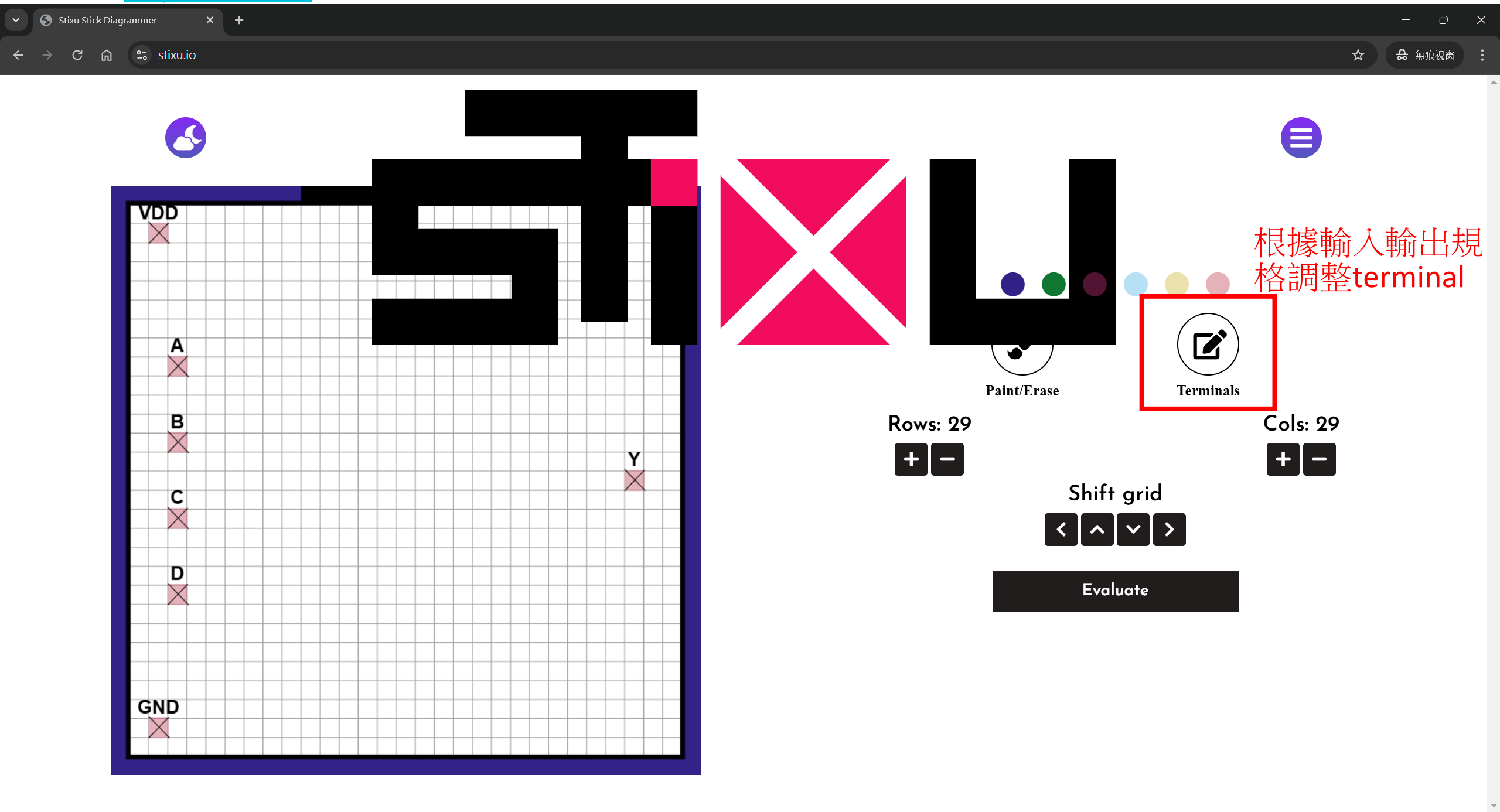
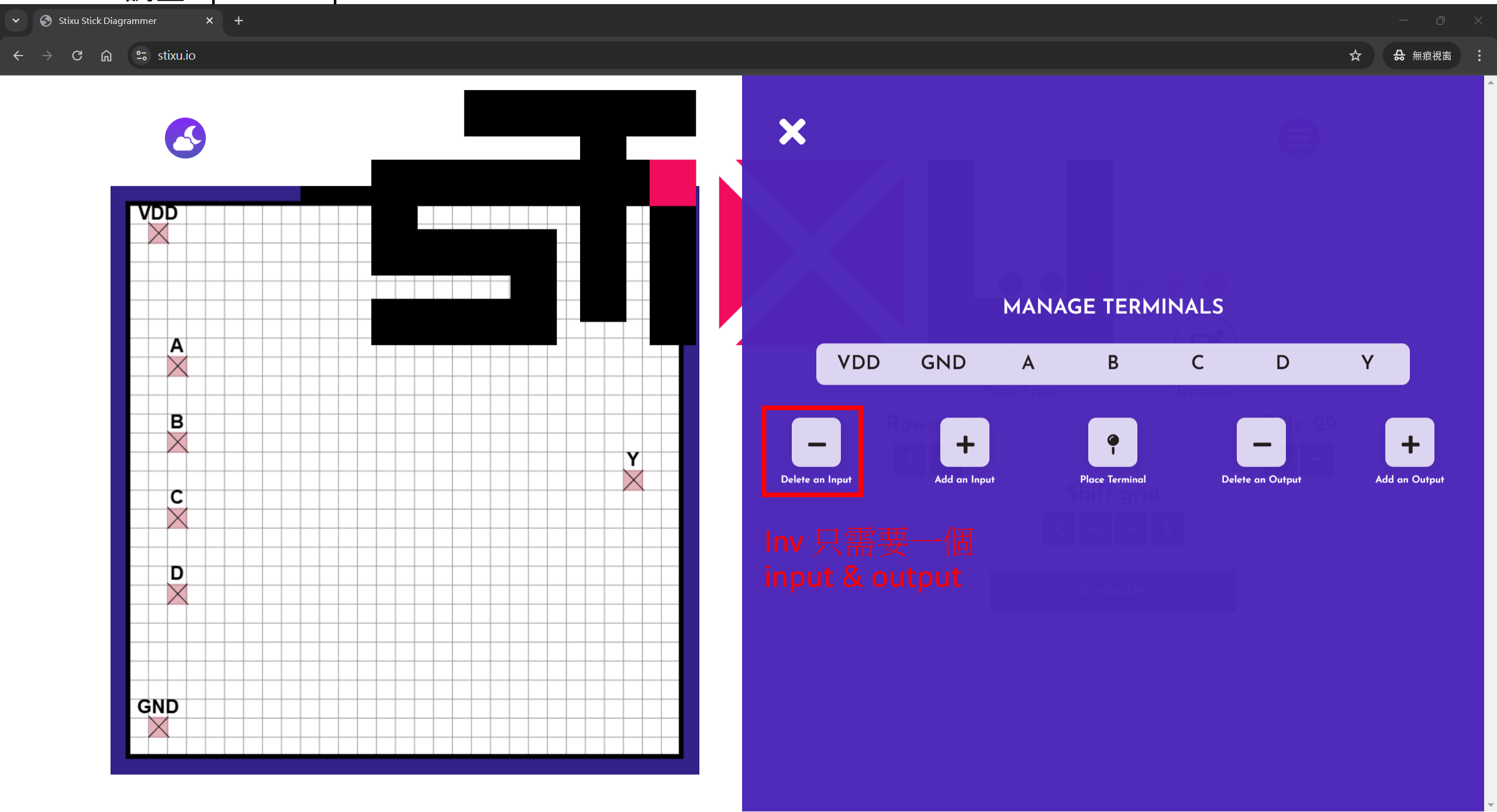
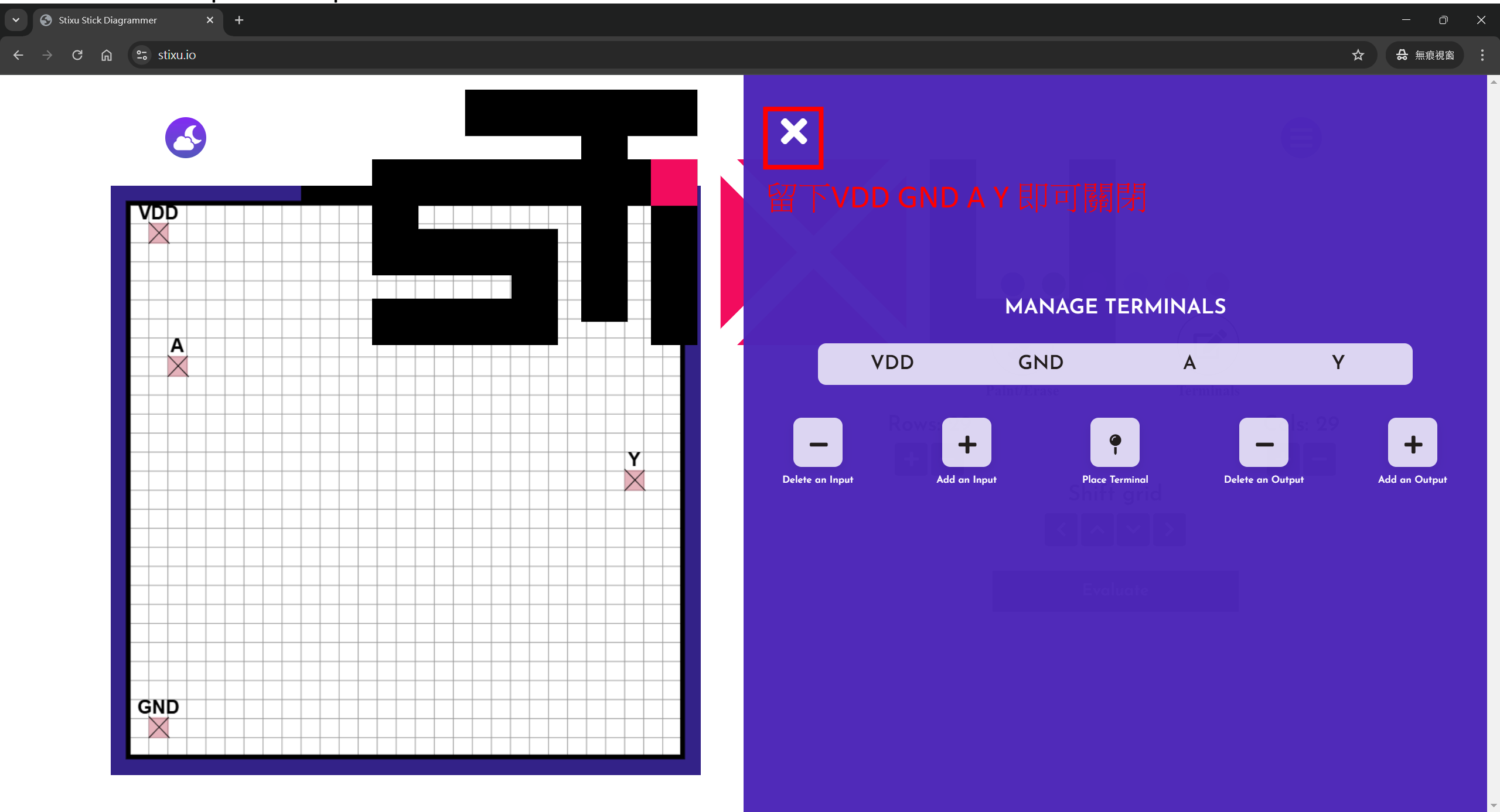
STEP 1 https://stixu.io inveter 畫法示範



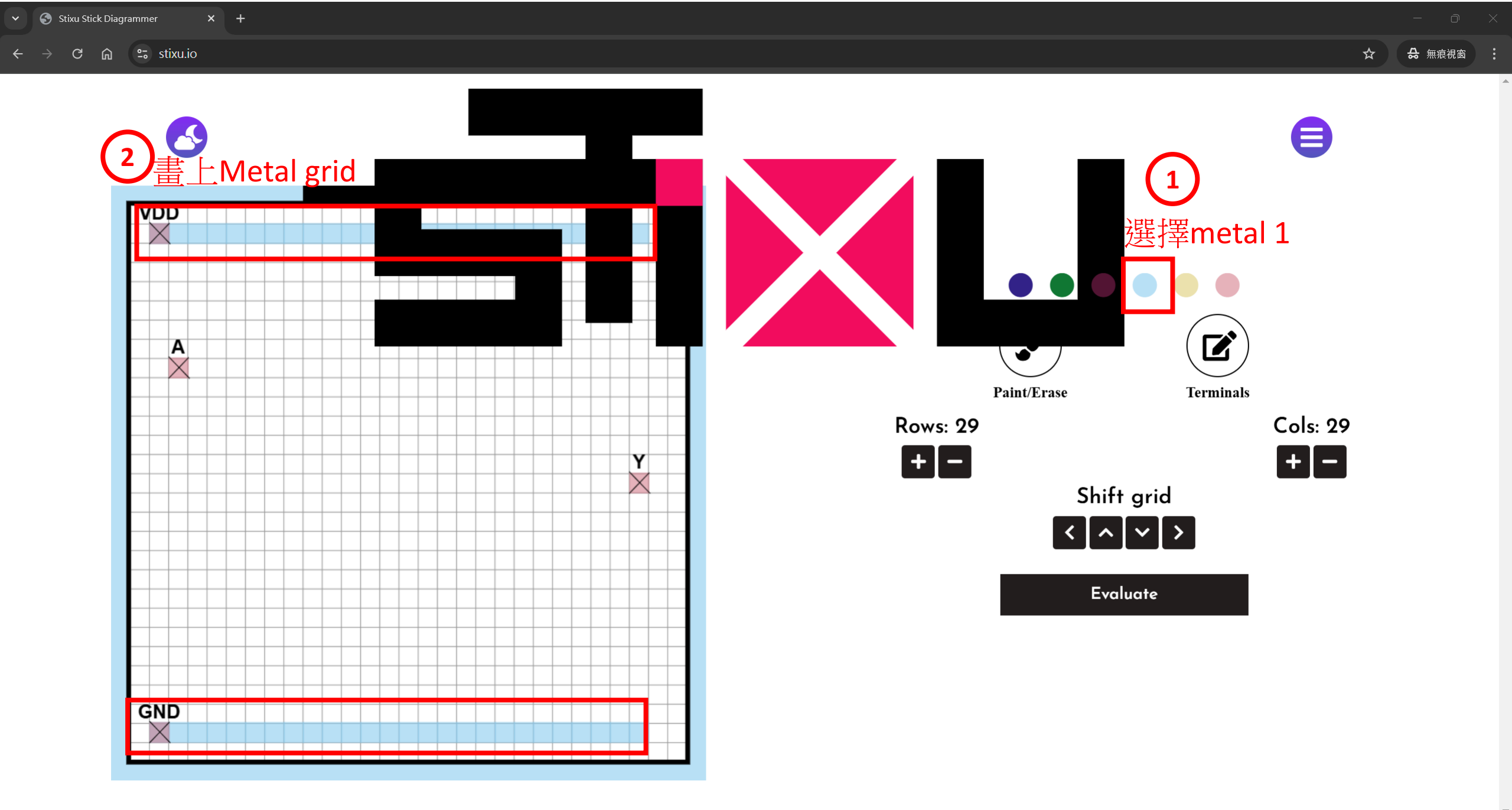
STEP 2　調整input & output



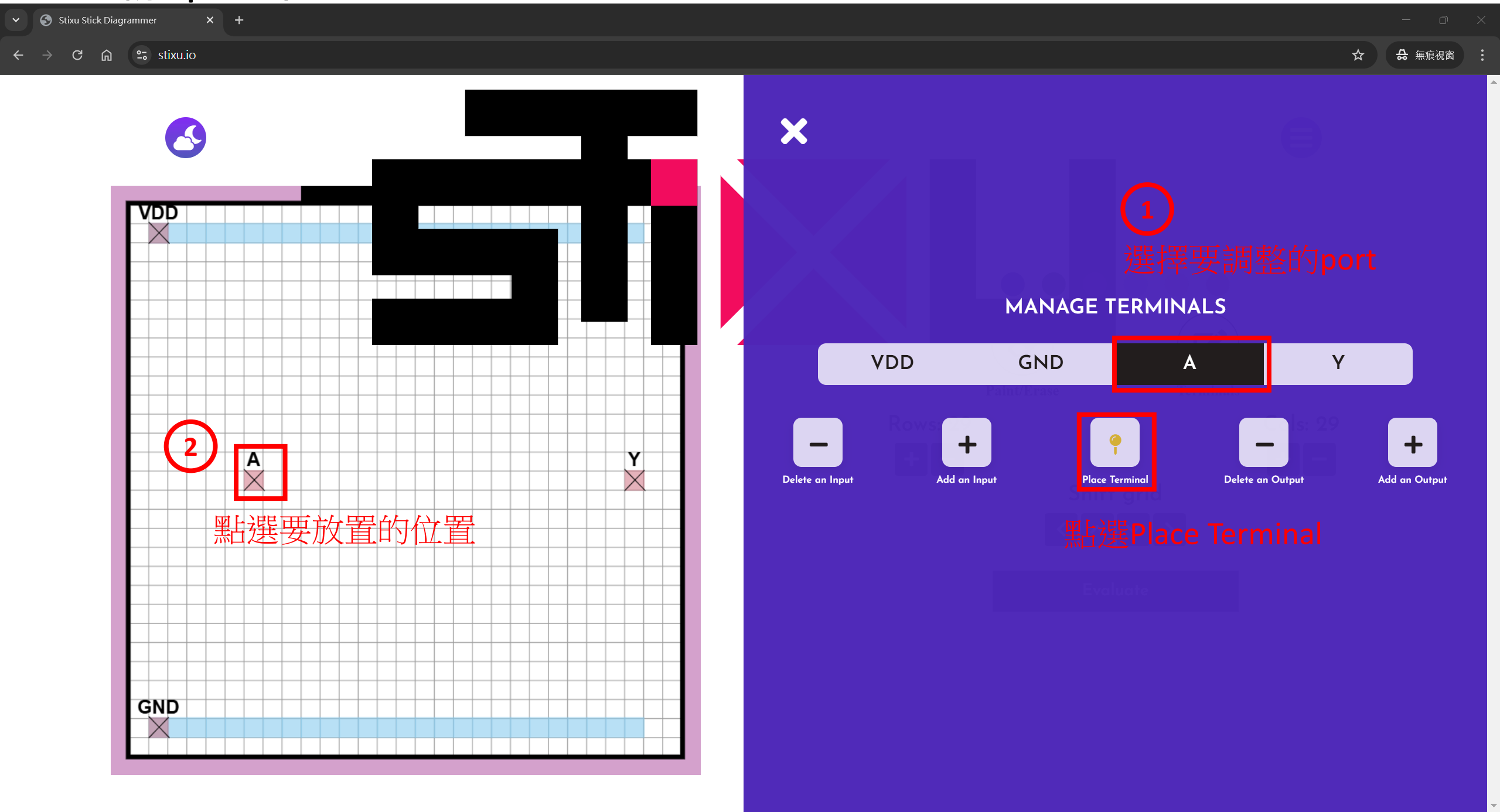
STEP 2　調整input & output



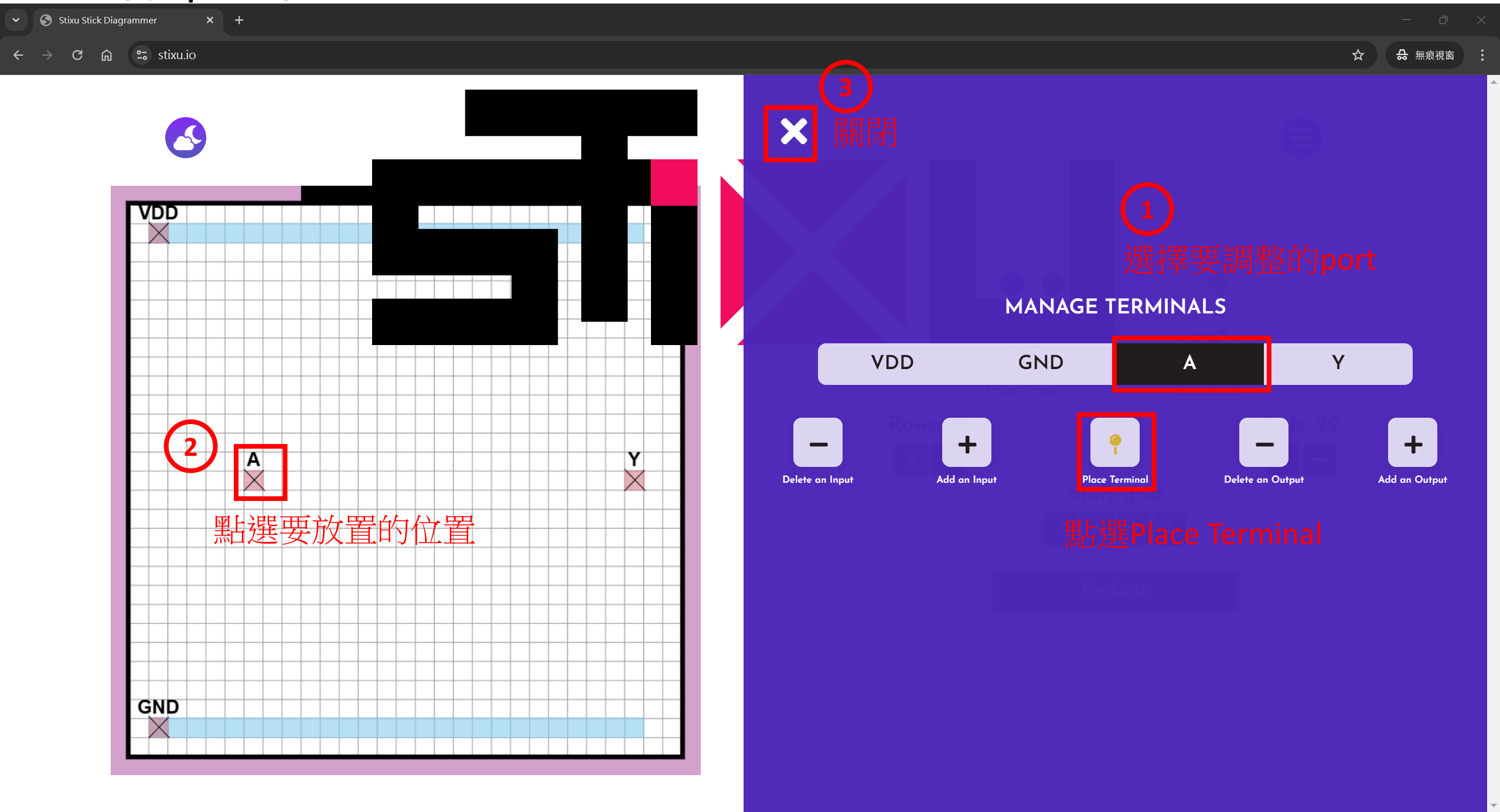
STEP 3　畫Metal grid



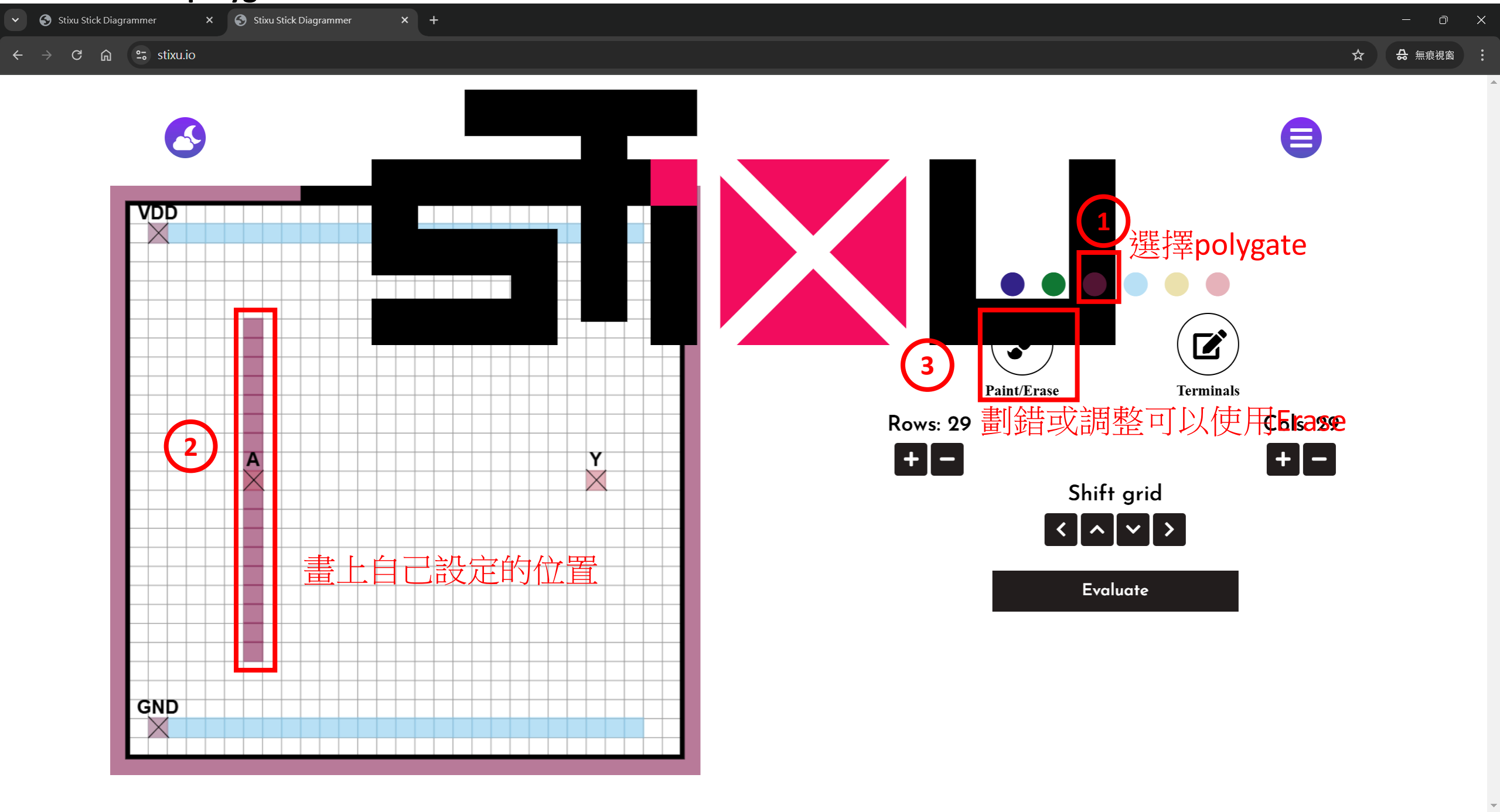
STEP 4　調整port 位置



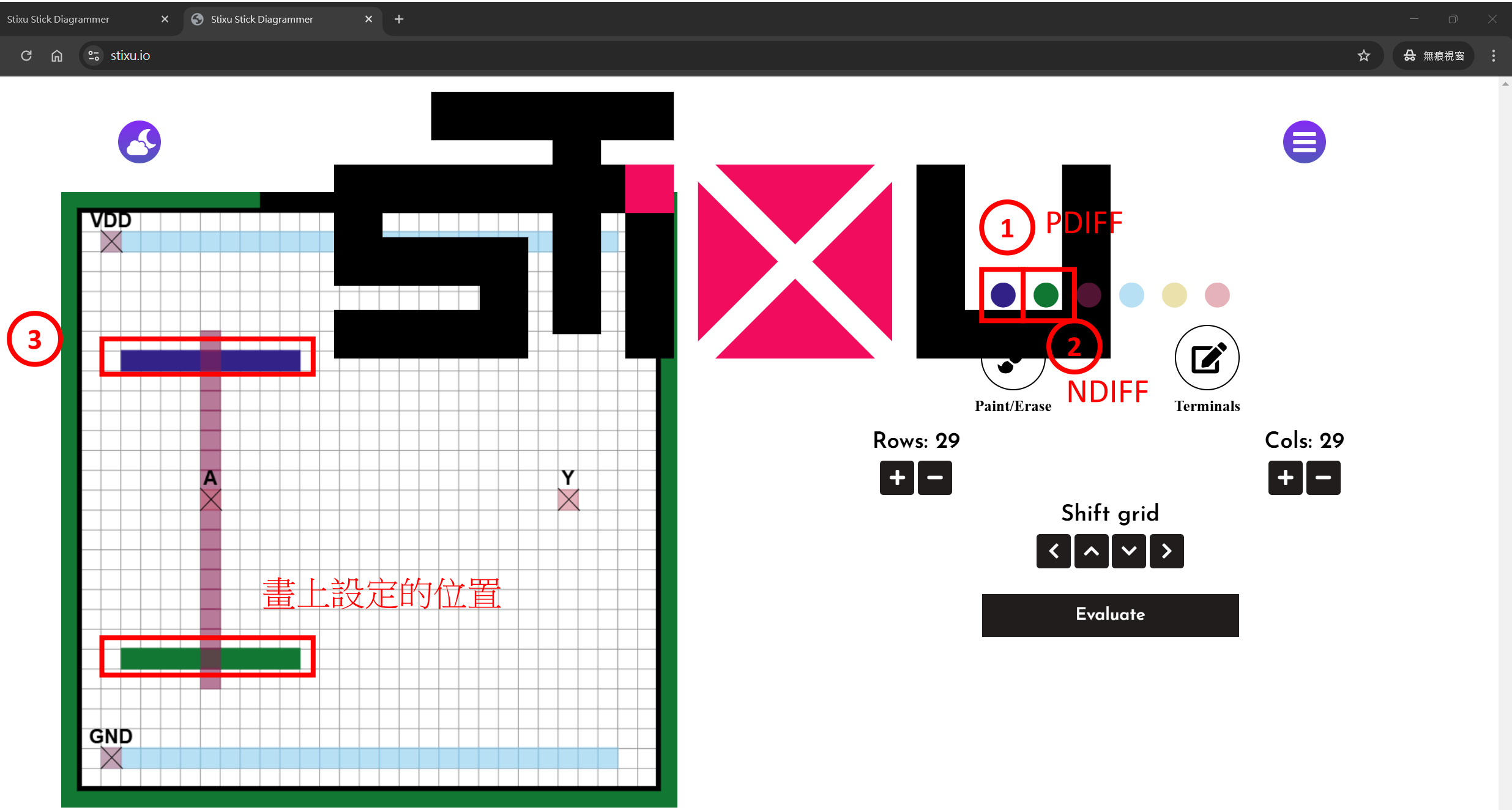
STEP 4　調整port 位置



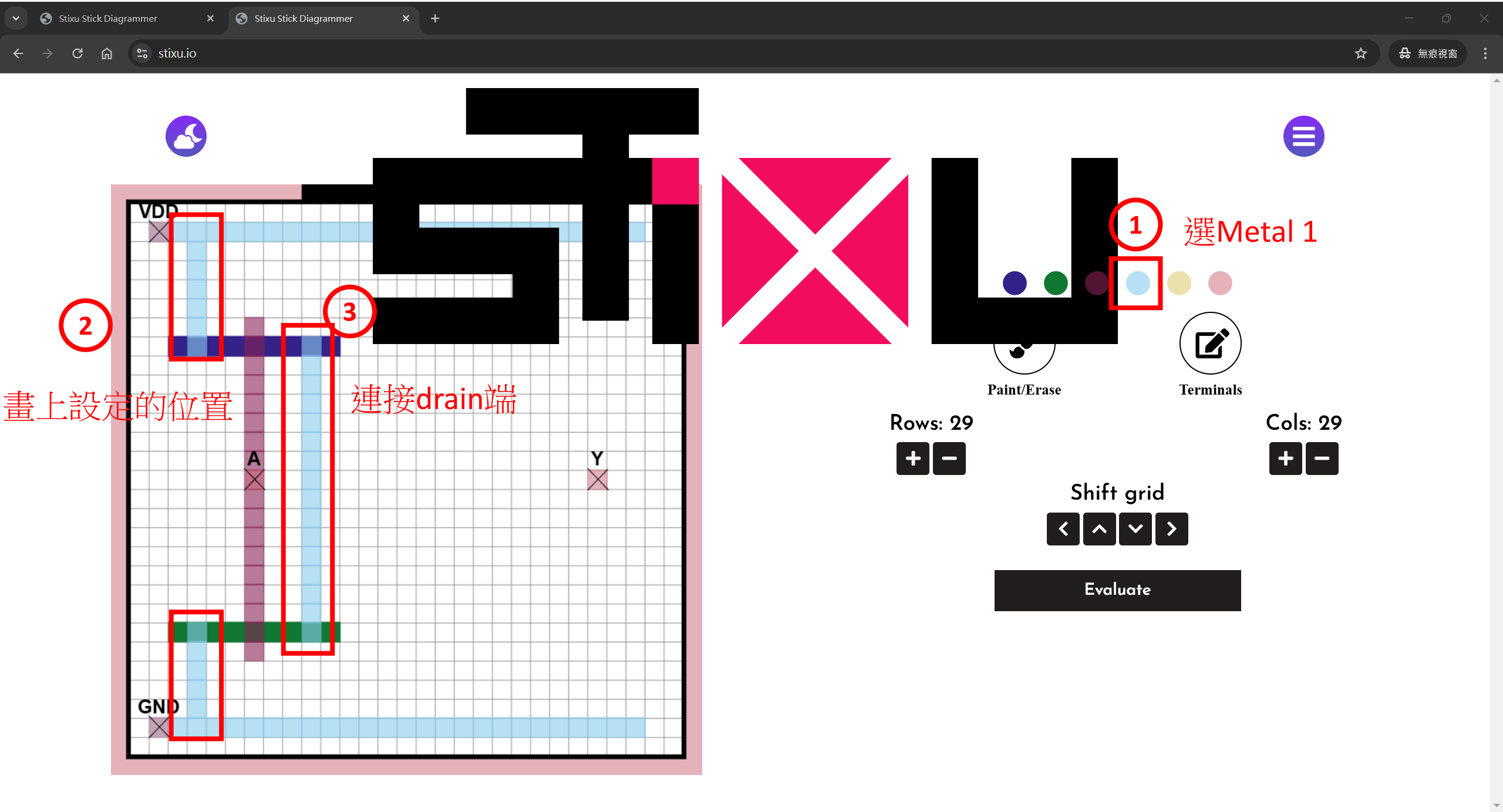
STEP 5　畫上polygate



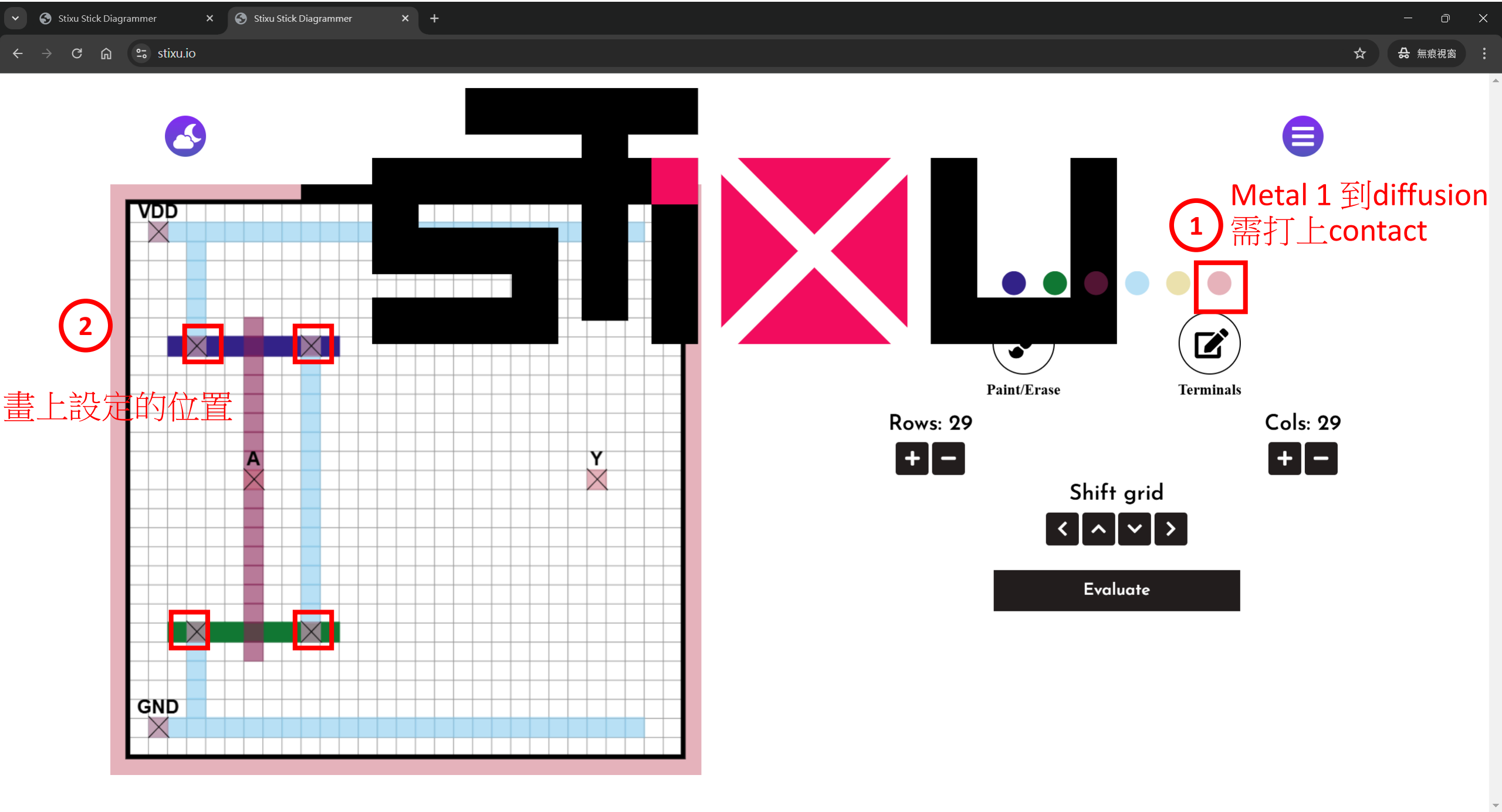
STEP 6　畫上P-diffusion & N-diffusion



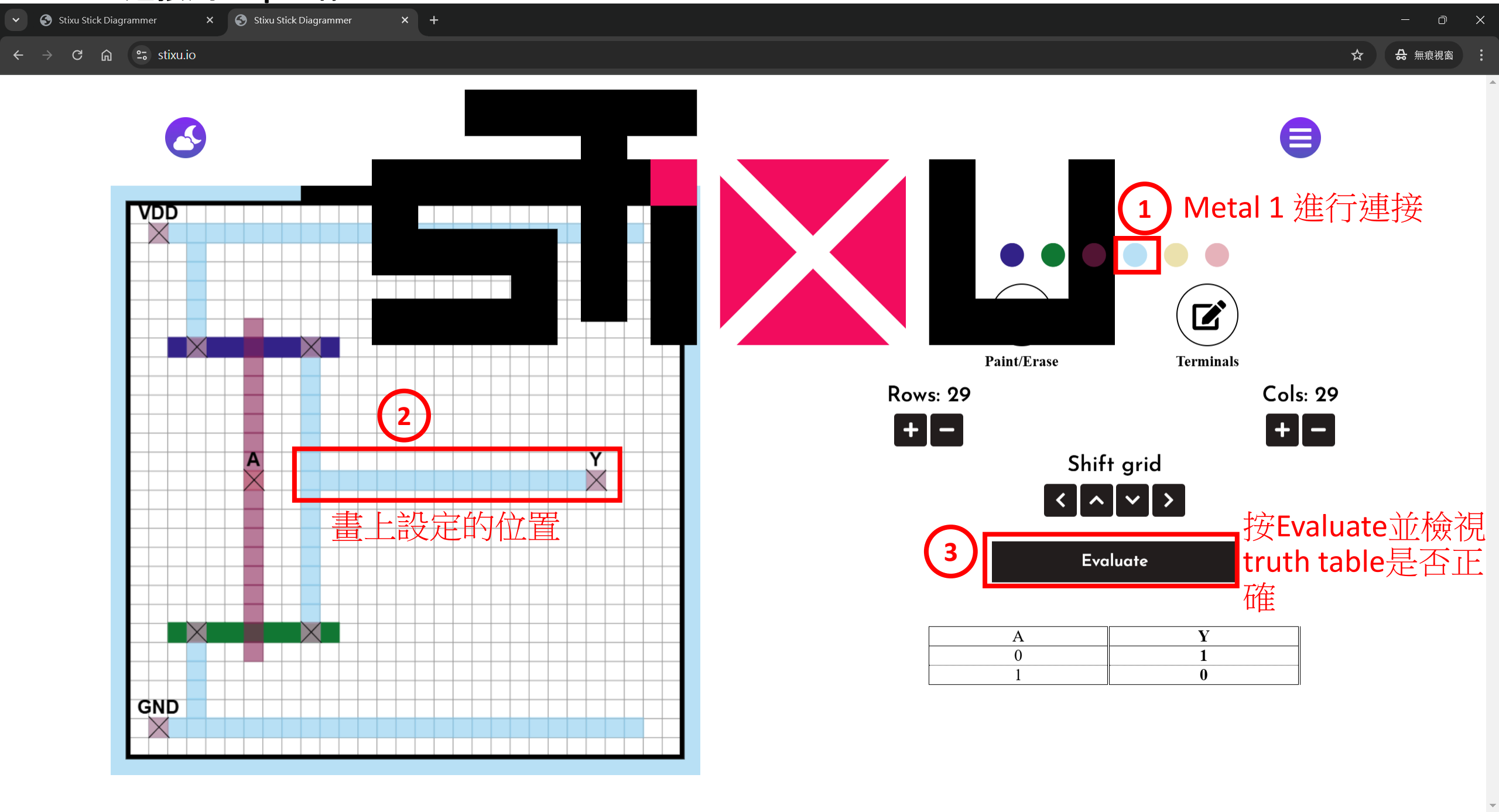
STEP 7　畫上連接VDD & GND 到MOS



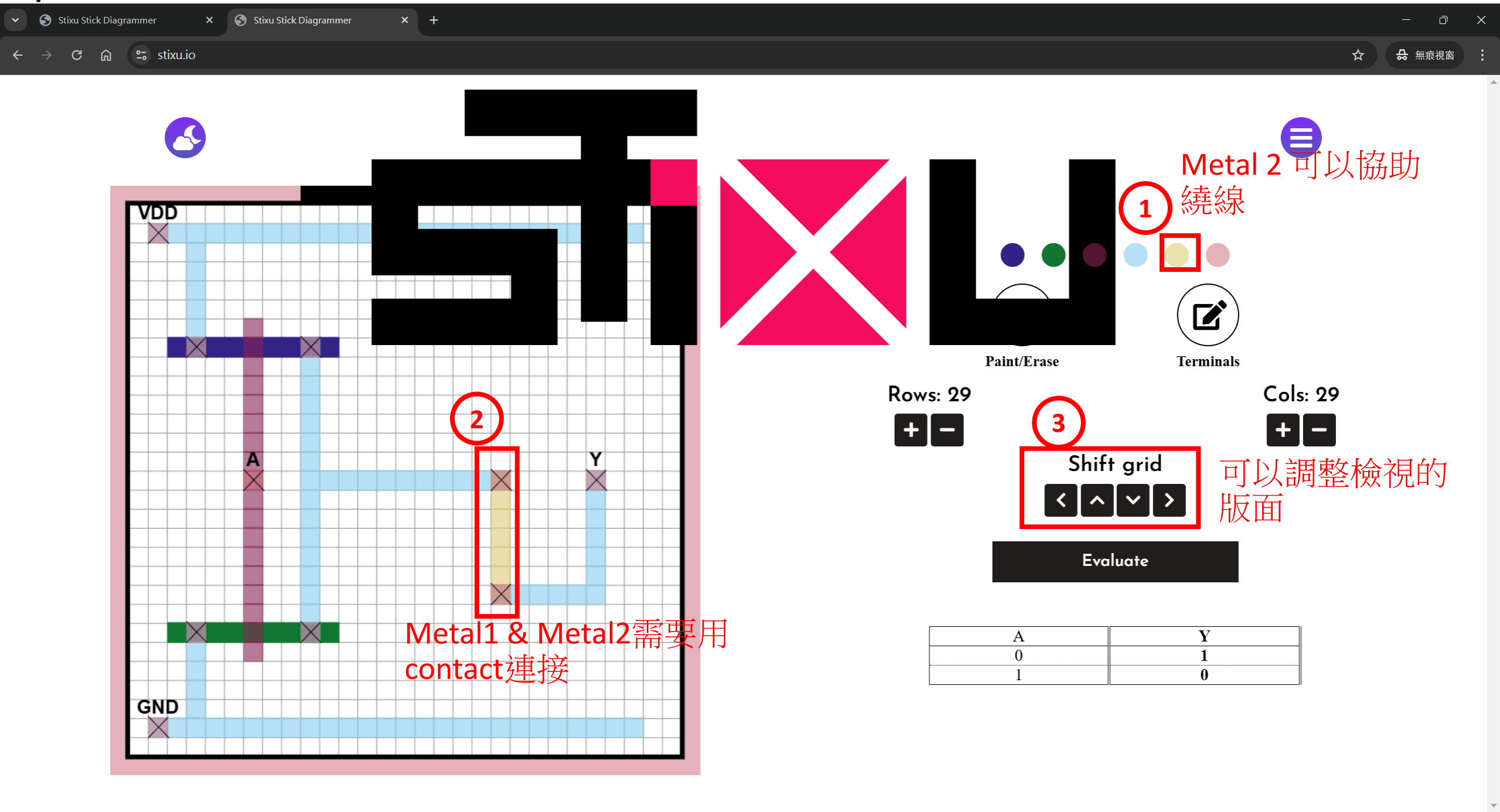
STEP 7　畫上連接VDD & GND 到MOS



STEP 8　連接到output端



Tips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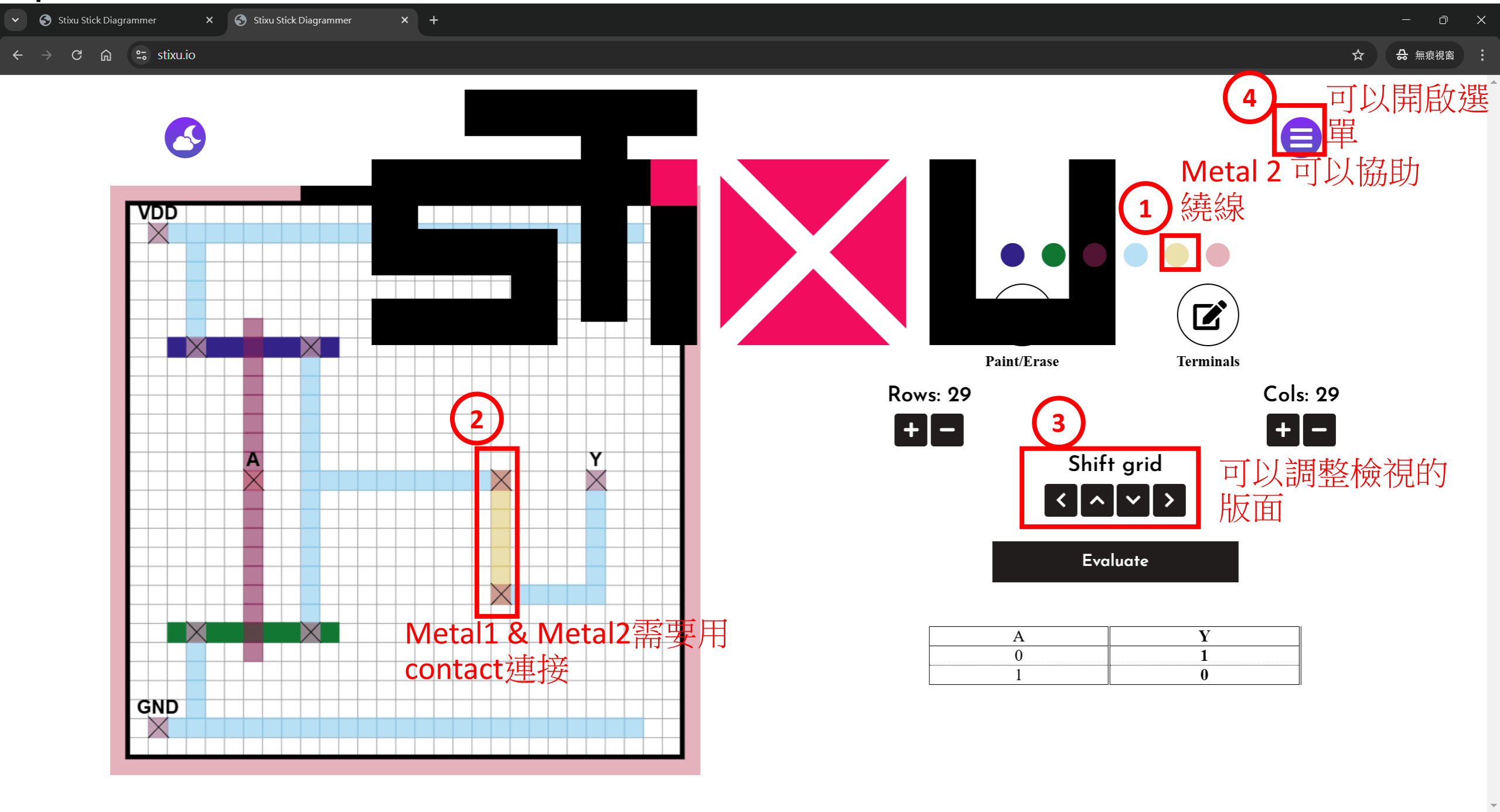
無痕視窗
VDD
A
Y
GND
1
2
3
Metal 2 可以協助繞線
Metal1 & Metal2需要用contact連接
Paint/Erase
Terminals
Rows: 29
Cols: 29
Shift grid
可以調整檢視的版面
Evaluate
A
Y
0
1
1
0

Tips


Stixu Stick Diagrammer
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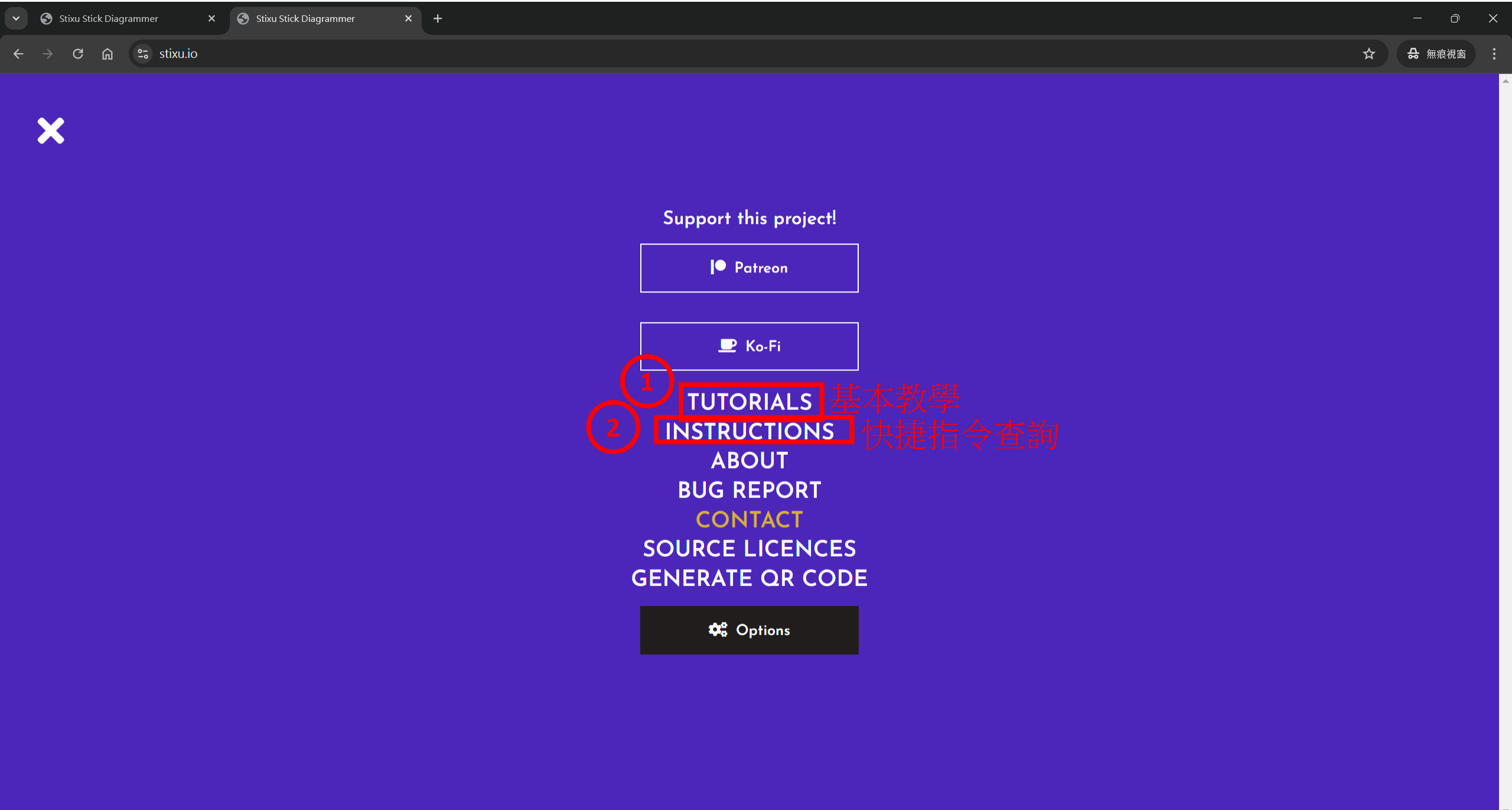
無痕視窗
VDD
A
Y
GND
2
Metal1 & Metal2需要用contact連接
1
Metal 2 可以協助繞線
4
可以開啟選單
Paint/Erase
Terminals
Rows: 29
Cols: 29
3
Shift grid
可以調整檢視的版面
Evaluate
A
Y
0
1
1
0

Tips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1
TUTORIALS
基本教學
2
INSTRUCTIONS
快捷指令查詢
ABOUT
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CONTACT
SOURCE LICENCES
GENERATE QR CODE
Options

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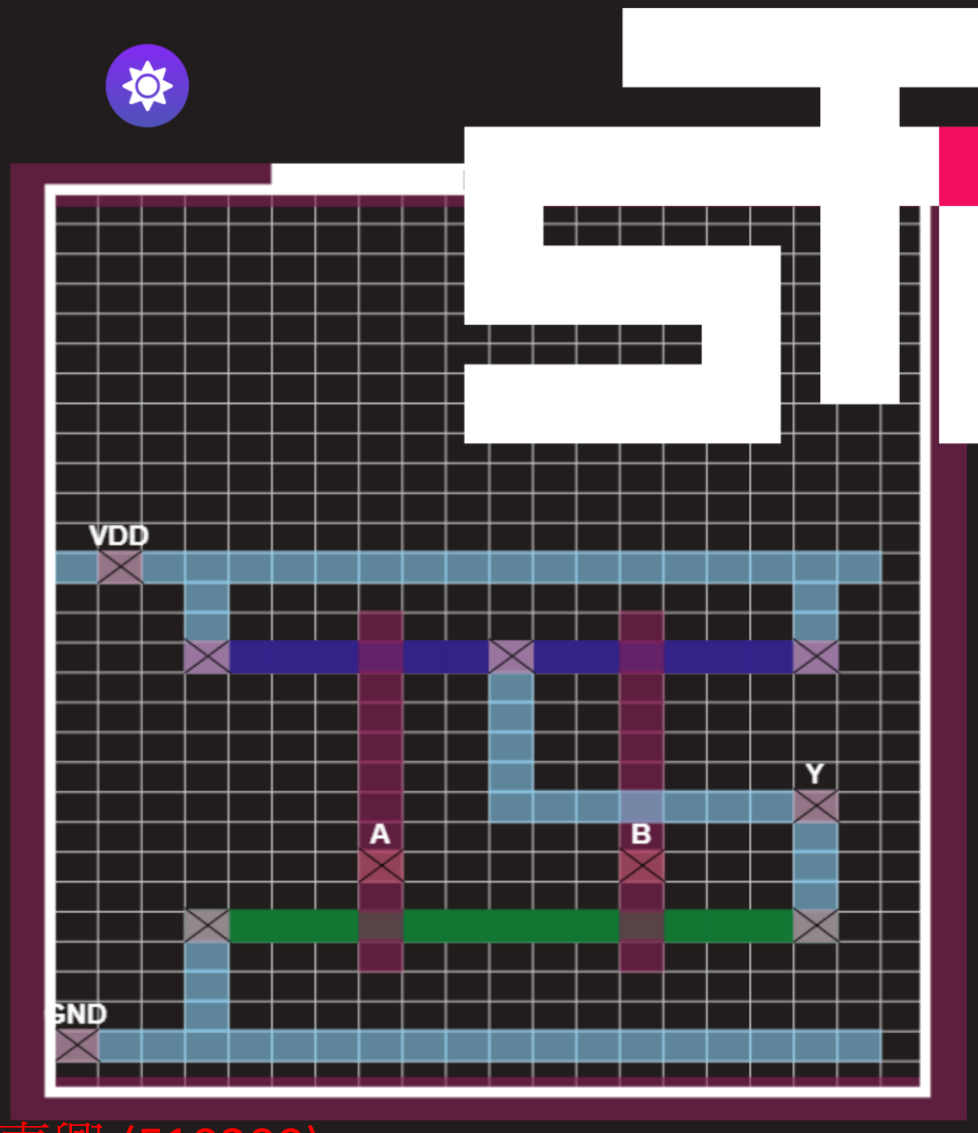

Paint/Erase

Terminals

Rows: 30

Cols: 20

Shift grid

Evaluate

NAND2

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

By 林嘉興 (F10300)